End-to-End Data Analytics Pipeline and Insights from the LEGO Catalog (1949–2026)

Alex Taschuk Jimi Ademola Sam Sikora

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1. Introduction

The LEGO catalog offers a unique lens through which to view changes in global desires and the use of licensed IP. This report outlines our methodology for extracting and cleaning this data, followed by a comprehensive analysis of how the brand has expanded its portfolio to cater to the "Adult Fans of LEGO" while maintaining its core appeal.

2. Data Pipeline and Justifications

To ensure the integrity of our insights, we established a robust pipeline to move raw data from external sources into a structured SQL environment.

2.1 Extraction and Preparation

The primary data source was the Rebrickable CSV catalog. During the extraction phase, we chose to convert the data into **Tab-Separated Values (TSV)**. This was a critical design choice to avoid delimiter conflicts, as LEGO set names often contain commas (e.g., "Police Station, 1970s") which can inadvertently break standard CSV parsing.

2.2 Cleaning and Bulk Loading

Data cleaning focused on type-casting for SQL compatibility. We mapped Boolean values to **0/1** types and ensured that empty strings were converted to **NULL** to maintain statistical accuracy during aggregation. For the ingestion phase, we utilized the **BCP (Bulk Copy Program)** utility. This allowed for high-performance loading that significantly outperformed standard `INSERT` statements.

2.3 Schema Design

The relational schema was designed with an emphasis on inventory relationships. The `inventories` table serves as the primary bridge between the `sets` and `parts` tables. This many-to-many relationship allows us to drill down from high-level theme trends to the specific elements used within individual sets.

3. Analysis and Key Conclusions

3.1 Set Complexity and Growth

The most significant trend observed is the dramatic increase in set complexity. For the first 50 years of production, the median piece count remained relatively static, staying below 65 pieces until the late 1990s. However, since 2020, the median has risen to approximately 207 pieces.

This growth, visualized in Figures 1 and 2, reflects LEGO's pivot toward adult consumers. Because piece counts are highly right-skewed, we utilized a log-scale transformation to better understand the distribution:

$$\log_{10}(\text{num_parts})$$

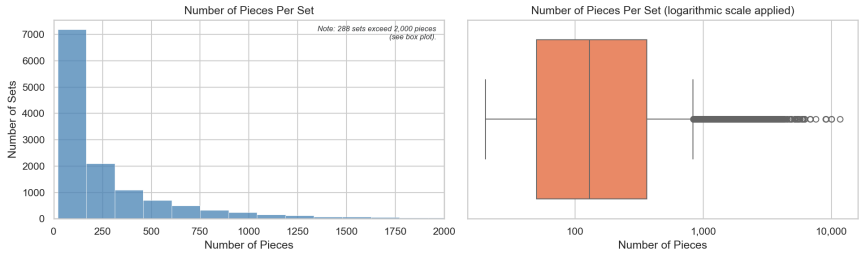

Figure 1: The distribution of piece counts per set remains heavily right-skewed.

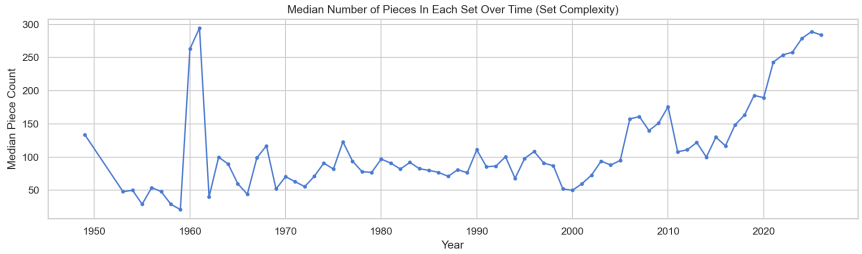

Figure 2: The median piece count demonstrates a sharp upward trajectory beginning in the late 1990s.

3.2 The Shift Toward Licensed Themes

LEGO has increasingly leaned into external Intellectual Property (IP). Licensed sets now account for over 40% of the portfolio, representing a move toward "collectible" sets that leverage existing fanbases like Star Wars.

3.3 Evolution of the Color Palette

The "look and feel" of LEGO has evolved from primary colors to a more realistic palette. Neutral tones now dominate, and transparent pieces account for 16.7% of the total catalog, reflecting the need for real-world detail.

4. Tableau Dashboards

The following interactive dashboards were developed in Tableau to complement the SQL-based analysis. Each dashboard targets a distinct analytical dimension, enabling dynamic filtering and drill-down exploration beyond what static figures can convey.

4.1 Set Complexity Over Time

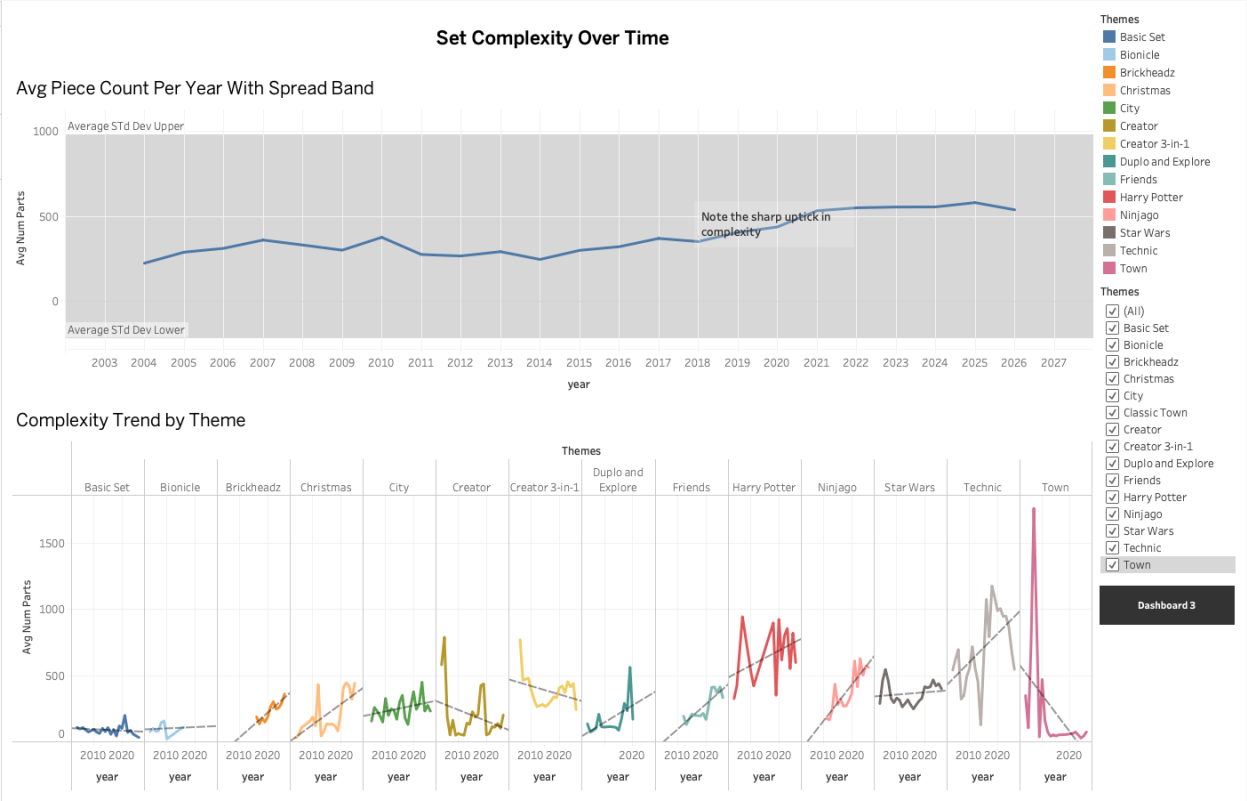

Figure 3: Tableau dashboard illustrating the evolution of set complexity (piece count) across decades. Filters allow exploration by theme and year range, highlighting the post-1990s surge in large-format sets.

4.2 Licensed vs. In-House Themes

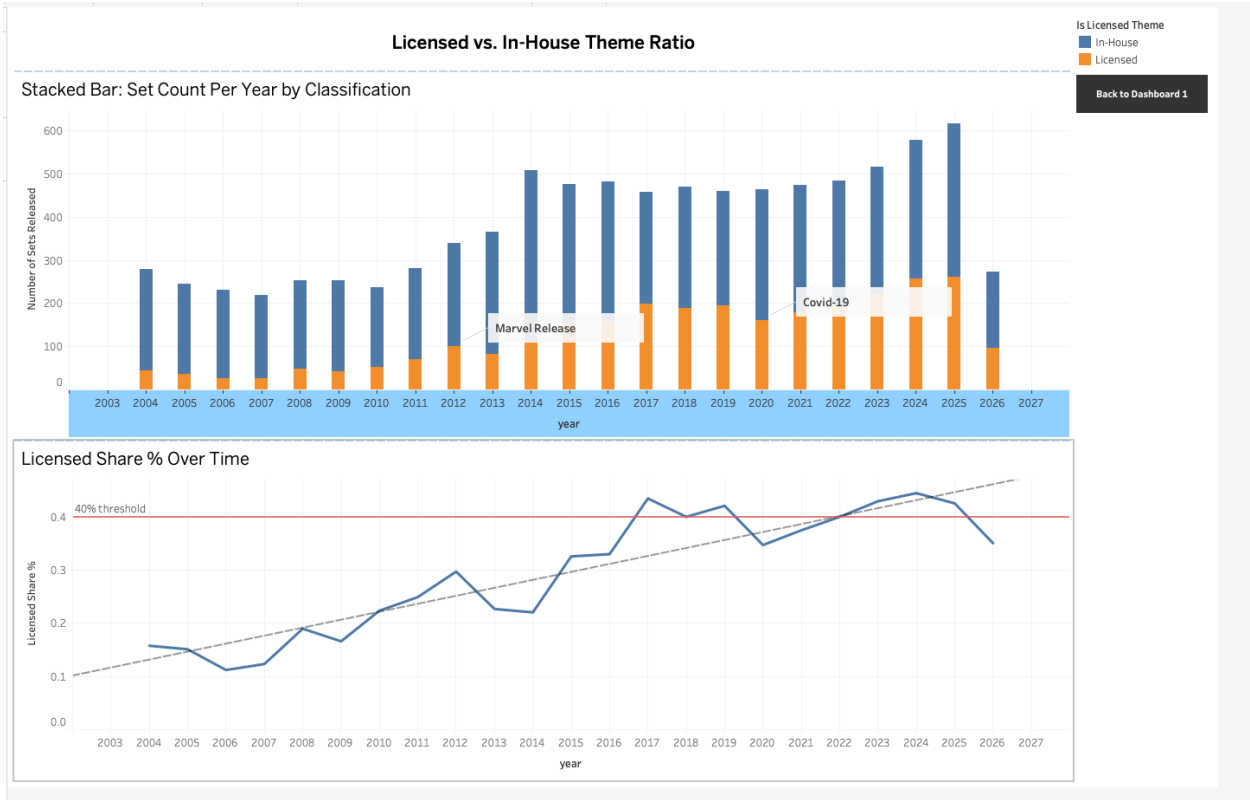

Figure 4: Tableau dashboard comparing the growth of licensed IP themes against in-house LEGO themes over time. The view surfaces the accelerating reliance on external franchises beginning in the late 1990s.

4.3 Colour Palette Evolution

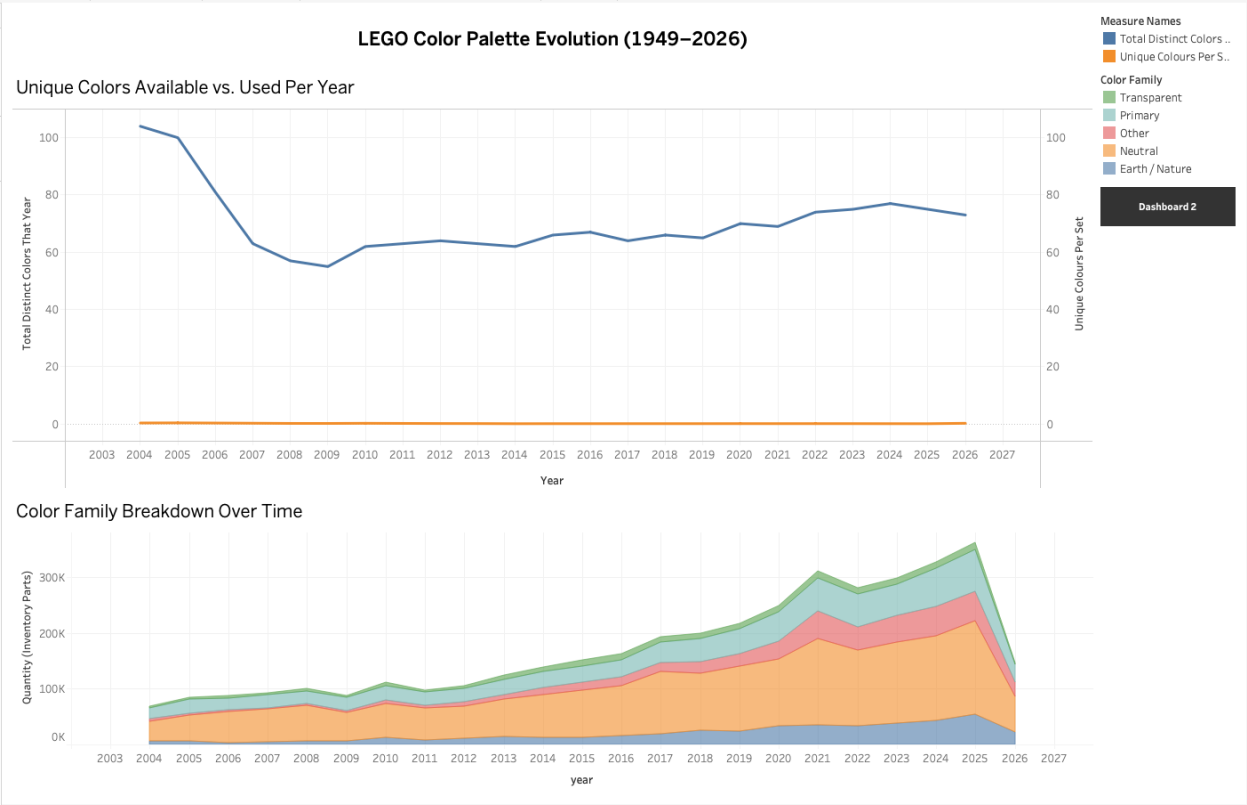

Figure 5: Tableau dashboard tracing the shift in LEGO's colour palette from primary tones to a broader, more naturalistic range. Transparent and neutral colours are highlighted to reflect the brand's move toward display-oriented sets.

5. Data Limitations

Despite the depth of the Rebrickable dataset, certain limitations exist. First, there is no native flag to distinguish between **In-House vs. Licensed** themes. Second, while `num_parts` is a strong proxy for complexity, it does not account for the **build difficulty** or the physical size of individual elements.

6. Final Summary

The data confirms that LEGO has successfully transitioned from a toy company to a collection and design business. The increase in piece counts, the reliance on high-value licenses, and the shift toward sophisticated / neutral color palettes point to a product line designed as much for display as for kids.